

Binary Search : Key not Found

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[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]
11	22	33	44	55	66	77	88	99

LEFT

MID

RIGHT

1) Key : 37

2) Left : 0
Right : 8
Mid : 4

3) Compare :
37 == 55 ? No

4) 37 is smaller to 55 , we will search in the Left sub array
Left sub-array : Left to Mid-1

[0]	[1]	[2]	[3]
11	22	33	44

2) Left : 0
Right : Mid-1 : 3
Mid : 1

3) Compare the key : 37 == 22 ? No

4) 37 is greater to 22, hence continue the search in RSA

[2]	[3]
33	44

RSA is from Mid+1 index to Right Index

Index 2 to index 3

2) Left : mid+1 : 2
Right : remains same as above iteration : 3
Mid : (2+3) / 2 : 2
3) 37 == 33 ? No
4) As 37 is greater to 33 we will search in RSA
RSA is from mid+1 to Right
2+1 to right
3 to 3

[3]
44

2) Left : 3
Right : 3
Mid : 3

3) Compare 37 == 44 ? No
4) Check as 37 is smaller to 44 continue your search in the left sub-array

the left sub-array

LSA is from Left to Mid-1

Left : 3

Right : mid - 1 : 3-1 : 2

As left crosses right variable we can say that the array is over for searching and the key is not found.